

Spring Framework

Core Spring ecosystem: IoC/DI, Web, data, testing and integration.

Fundamentals

Spring Framework provides infrastructure for Java applications including dependency injection, configuration, Web MVC/WebFlux, data access and testing support.

Official site and documentation

Use the official Spring Framework source as the primary reference:

<https://spring.io/projects/spring-framework>. Official documentation should be the first source for installation, supported versions, syntax changes and new features.

Why QA should know it

Understanding controllers, services, repositories, beans and profiles helps QA locate failures and collaborate with developers.

Projects and dependencies

Modern projects usually consume Spring through Maven/Gradle and often through Spring Boot. The framework is not installed as a standalone desktop application.

Testing

The ecosystem supports context, integration and web testing. QA Automation can combine JUnit/Mockito with Spring testing capabilities.

Configuration

Learn properties/YAML, profiles, environment variables and dependencies because they strongly affect QA environments.

When to use it

Use this knowledge when product risk, architecture or the delivery stage requires this type of validation. Confirm the goal, environment, required data and expected evidence before starting.

How to start safely

Start from the official Spring Framework site and documentation. Confirm prerequisites, supported versions, operating system and required permissions. Learn in a controlled environment before using it in a critical workflow.

First QA-oriented use

Pick one small, verifiable task in Spring Framework, configure only what is necessary, run a simple example and confirm the outcome. Repeat with different data or conditions before adding abstractions.

Using it in a real project

Define where Spring Framework fits, who owns configuration, how credentials and data are handled and how results are shared. Integrate it into the team workflow instead of creating an isolated activity.

Adoption good practices

Standardize versions and configuration, document important decisions, keep secrets out of code and preserve reproducible examples. Reassess whether Spring Framework still delivers value as the product evolves.

Common issues and troubleshooting

When Spring Framework fails, verify versions, dependencies, permissions, connectivity, data and environment before assuming a product defect. Preserve the original error and reduce the issue to the smallest reproducible case.